Breast Lesion Detection Using Weakly Dependent Customized Features and Machine Learning Models with Explainable Artificial Intelligence

1. Executive Summary

This report documents an independent implementation and critical methodological audit of the weakly dependent feature approach proposed by Moldovanu et al. (2025) for BUSI ultrasound images. It does not claim exact numerical reproduction of the paper's Table 3.

The study contributes (1) an independent extraction and algebraic validation of bounded-histogram (CHi) and grayscale-density (Ci) features from ground-truth lesion masks; (2) an audit of the authors' publicly released feature tables, which reveals complete cross-table label disagreement across the matched feature vectors and numerical behavior inconsistent with normalized density ratios; and (3) an extension that combines density ratios with mask-derived geometric descriptors and lesion-intensity statistics.

The primary model registry excludes two records associated with an exact pixel-identical cross-class image conflict. On the resulting 645-sample registry, the extended feature family achieves higher holdout accuracy than the density-ratio baseline for all four evaluated classifiers. Paired exact McNemar tests with Holm adjustment support this difference on the current 162-sample holdout. These findings are preliminary and remain conditional on sample-level evaluation rather than independent patient-level validation.

The study therefore constitutes an independent reimplementation, critical data audit, and methodological extension. It does not establish exact numerical reproduction, external clinical validity, or readiness for clinical deployment.

2. Project Positioning

Claim level: independent reimplementation + methodological extension + critical data audit.

Not claimed: exact reproduction of published accuracies, identical sample selection, patient-level independence, or clinical deployment readiness.

This framing matters. The paper reports very high Random Forest accuracy (0.969 in Table 3; the discussion also mentions 100% in places), but provides no training code and only two feature CSVs. Under those conditions, an honest independent study is more valuable than a forced numerical match.

3. Reference Paper — What Was Claimed

Moldovanu et al. propose two weakly dependent feature classes extracted from BUSI lesion masks:

- CHi — bounded histogram features: eight non-overlapping 32-level gray bins over the ROI.
- Ci — grayscale density features: area of each bin region divided by ground-truth mask area.

Classifiers include Random Forest, XGBoost, Gradient Boosting, and an L1-regularized linear model referred to as LASSO. Evaluation uses a 75:25 split and five-fold cross-validation. Explainability relies on LIME and SHAP. The public artifacts are limited to https://github.com/simonamoldovanu/features_breast (two CSV files) plus a MATLAB R2021a note for feature extraction.

Table 3 reports, among other values, RF accuracy 0.969 for both feature families, with lower scores for XGB and the linear model. The discussion text is not fully consistent with that table when it refers to 100% RF accuracy.

4. Data and Integrity Findings

4.1 BUSI inventory

The local Dataset_BUSI_with_GT inventory used in this study contains 437 benign and 210 malignant original images with usable masks (paper: 438 and 211). No synthetic samples were added. The one-sample-per-class shortfall is documented in the dataset reconciliation audit and treated as a version/selection difference, not as a defect to be papered over.

4.2 Cross-class pixel duplicate

Image-integrity analysis identified at least one exact pixel-identical pair carrying opposite labels (benign versus malignant). That conflict is methodologically serious: the same visual content cannot honestly support both classes. Cell 10B quarantines such pairs from the primary registry while keeping an unfiltered copy for sensitivity analysis. Primary downstream metrics are therefore computed on the cleaned set.

4.3 Authors' feature tables

The two publicly released feature tables contain 647 rows each. Their eight-dimensional numerical feature vectors have identical multisets, while all 647 matched vector occurrences have opposite mapped class labels. In addition, the table named density_features.csv contains large integer-valued features rather than rows behaving as normalized density ratios.

These observations establish a cross-table consistency problem, but they do not identify which file contains the correct label for an individual image. The article describes 649 selected benign and malignant images, whereas the public CSV tables contain 647 rows. The local BUSI inventory also contains 647 original images across these two classes, compared with the 649 samples reported in the article.

The author-derived tables are therefore retained for provenance and audit purposes but excluded from model training. No labels are automatically reversed, and no exact correspondence between the author's rows and the local image files is assumed.

5. Independent Feature Pipeline

Features were recomputed from BUSI images and ground-truth masks:

- CH1–CH8: raw pixel counts in bins [0–31], [32–63], …, [224–255] inside the mask.
- C1–C8: Ci = CHi / mask_area_px.

Algebraic checks confirm that, for non-empty masks, the CH bins sum to the mask area and the C ratios sum to one (within floating-point tolerance). That validates internal consistency of the independent implementation. It does not prove numerical identity with the authors' MATLAB code, whose public description mixes normalized histogram language with pseudocode that is ambiguous about counts versus frequencies.

Three experiment families were registered:

- own_CH_raw — raw histogram counts only.
- own_C_ratio — density ratios only.
- own_CV_extended — density ratios plus geometric descriptors (mask_extent, bbox_aspect_ratio, lesion_area_fraction, and related shape cues).

6. Modelling and Evaluation Protocol

Estimators: Random Forest, XGBoost, Gradient Boosting Classifier, and L1-regularized logistic regression (saga). The last is deliberately not called "LASSO regression," because L1 logistic classification is a different estimator from ordinary Lasso.

Protocol elements that diverge from a pure clone of the paper are stated openly: nested evaluation with a stratified hold-out plus cross-validation inside training; invalid paper hyperparameters (for example, criterion on XGB) are not forced into APIs that do not support them; patient-level splits are impossible without patient IDs in the public BUSI release.

All primary metrics reported for the study should be taken from the cleaned registry after quarantine. Sensitivity runs on the unfiltered registry quantify how much the duplicate pair moves the numbers.

7. Explainability Stack

The notebook implements a fuller XAI stack than the paper's LIME + SHAP pair:

- Native tree importance where available.
- Permutation importance on hold-out data.
- SHAP (TreeSHAP when supported; model-agnostic fallback for XGB when categorical/split limitations appear).
- LIME local explanations for selected instances.
- Rank-agreement analysis (Spearman) between SHAP and permutation rankings.

With only eight or thirteen features, rank correlations and p-values are exploratory. They support qualitative discussion of stability across explainers; they should not be oversold as confirmatory statistics.

8. Main Result: Mask-Derived Feature Extension

The main positive result is the improvement associated with own_CV_extended, which combines the eight normalized histogram ratios with three geometric descriptors and two lesion-intensity statistics.

On the current 162-sample holdout, accuracy increases relative to the own_C_ratio baseline for each of the four evaluated classifiers. Paired exact McNemar tests, adjusted using the Holm procedure across the registered comparisons, support the observed difference in classification correctness for all four baseline-versus-extension comparisons.

This result suggests that adding mask-derived shape, extent, and intensity-statistic descriptors may provide useful complementary information beyond normalized intensity-bin ratios. It does not isolate the effect of geometry alone, because the extension adds both geometric and intensity-statistical information.

All extension features depend on a ground-truth lesion mask. The experiment consequently evaluates classification of mask-conditioned lesions, not end-to-end detection from unsegmented ultrasound images. External validation, patient-level independence, and clinical applicability remain unestablished.

9. Artifact and Audit Trail

The run produces a complete provenance package suitable for open science:

Artifact	Role
FINAL_PAPER_REPRODUCTION_AUDIT.md / .json	Human and machine-readable final audit; exact reproduction = False
primary_duplicate_quarantine_manifest_cell10b.json	Which samples were removed from the primary registry and why
feature_identity_audit.json	Algebraic consistency of independent CHi/Ci features
author_csv_reference_gate_audit.json	Why author CSVs were blocked from training
model_manifest.json (+ audit metadata)	Serialized models, checksums, round-trip status
dataset_inventory_* + fingerprint	BUSI counts and data fingerprint for the run
environment_*.json	Python and package versions; mismatch with paper stack recorded
csv/ + figures/	Metrics, McNemar, importance, SHAP/LIME plots

Environment note: the executed stack (e.g., Python 3.13-compatible constraints, scikit-learn 1.6.x, SHAP 0.52) is not the paper's Python 3.10 / scikit-learn 1.4.1 / SHAP 0.46 environment. Compatibility is recorded; exact environment cloning is not claimed.

10. Limitations

- Exact numerical reproduction of Table 3 was not established and is not asserted.

- Author feature tables are internally inconsistent; they cannot anchor a faithful replay of the published numbers.
- Sample-level stratified splits do not imply patient-level independence.
- Mask-based features are not equivalent to detection from unmasked images.
- XAI rankings with few features remain sensitive to sampling and method choice.
- Discussion text in the paper that cites 100% RF accuracy conflicts with Table 3's 0.969 and should not be treated as a target.

11. Recommendations for GitHub Publication

Publish under a title such as: Independent Reimplementation and Geometric Extension of Weakly Dependent BUSI Features (Critical Audit of Moldovanu et al., 2025).

Recommended repository contents:

- Notebook (or clean .py export) with the positioning statement at the top.
- FINAL audit Markdown/JSON and the quarantine, feature-identity, and author-gate JSON files.
- Key CSVs (metrics, McNemar, protocol audit) and a small set of figures.
- model_manifest.json; large .joblib weights via Git LFS or external link only.
- README that states clearly what is and is not claimed, highlights own_CV_extended, and summarizes the author-CSV problems.

Do not frame the repository as a successful exact reproduction. The audit trail already rejects that claim; leaning into independent science is both more accurate and more interesting.

12. Conclusion

The project succeeds as a careful independent study. It rebuilds the core CHi/Ci idea from public images and masks, exposes serious issues in the only feature data the authors released, and shows that a modest geometric extension can improve discrimination under paired testing. Those are substantive results. They do not require matching every digit in Table 3 to be worth sharing.

For readers and reviewers, the honest summary is: the reference paper's public materials are insufficient for exact reproduction; the independent pipeline is coherent; and extending intensity-derived densities with shape cues is a promising, testable direction on BUSI-style data.